

# India's Digital Revolution

## IT Services, UPI, and the Software Superpower That Built Itself

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### Executive Summary

India's technology sector has undergone a structural metamorphosis over the past three decades — from a low-cost back-office outsourcing destination for Western corporations to a full-stack technology ecosystem encompassing frontier AI research, semiconductor manufacturing ambitions, and the world's most sophisticated digital public infrastructure. As of August 2026, India's IT and technology exports exceed \$280 billion annually, the Unified Payments Interface (UPI) processes more than 131 billion transactions per year worth approximately \$2.3 trillion, and the Indian government's Semiconductor Mission has catalysed over \$15 billion in committed fab investments from Micron Technology, Tata Electronics, and the HCL-Foxconn joint venture. The arc of India's technology development — from the software services revolution of the 1990s to the semiconductor fabrication ambitions of the 2020s — represents one of the most consequential industrial transformations in the developing world.

India's technology ambition is simultaneously a commercial story, a geopolitical story, and a civilisational one. The same country that produces more software engineers than any other on earth is attempting to capture the upstream — chip design, semiconductor fabrication, advanced materials — that has historically been monopolised by the United States, Taiwan, South Korea, Japan, and the Netherlands. The gap between ambition and execution is real; India's semiconductor programme is nascent compared to TSMC's 30-year head start. But the scale of the bet — government incentives exceeding \$15 billion, the combined manufacturing depth of Tata Group and global partners, and a domestic market of 1.44 billion people — means India's technology trajectory will be one of the defining industrial stories of the next decade.

### Part I — The IT Services Colossus: \$280 Billion and Counting

#### The Software Services Foundation

India's IT and business process management (IT-BPM) industry generated approximately \$282 billion in revenue in FY2026 — a number that would make it the 8th-largest economy in the world if it were a standalone country. The industry employs approximately 5.8 million professionals directly and supports an additional 15 million indirect jobs across services, real estate, and infrastructure. The four largest Indian IT companies — Tata Consultancy Services (TCS), Infosys, Wipro, and HCL Technologies — together account for approximately \$73 billion in annual revenue and are among the most valuable corporations in Asia.

The IT industry's geographic concentration — Bengaluru ("India's Silicon Valley"), Hyderabad, Pune, Chennai, and NCR (Delhi/Gurugram/Noida) — has produced regional technology clusters with deep talent pipelines, infrastructure ecosystems, and venture capital networks. Bengaluru alone hosts the

Indian operations of over 400 Fortune 500 companies and has produced more technology unicorns per capita than any other Asian city outside of Shanghai and Beijing.

### **AI and the Service Industry Disruption**

The emergence of generative AI — particularly large language model (LLM) deployment for code generation, document processing, and customer service automation — presents the most significant structural threat to India's IT services industry since the Y2K-driven outsourcing boom created it. Infosys CEO Salil Parekh stated in Q1 FY2027 results commentary that "AI is fundamentally restructuring our project composition" — revenue from traditional application maintenance, testing, and BPO voice services (the historical bread and butter of Indian IT) is under pricing pressure as AI tools automate tasks that previously required human teams of 50-200 people.

Indian IT companies are responding by investing aggressively in AI capability: TCS's AI Cloud platform, Infosys Topaz (generative AI suite), and Wipro's ai360 are all attempts to move the Indian IT value proposition from "labour cost arbitrage" to "AI-augmented services at scale." The fundamental question for the next decade: can India's IT majors capture enough of the AI value chain to maintain their revenue and margin profiles as commoditised services compress? The early evidence (FY2025-2026 revenue growth rates of 4-6%, below the 12-15% rates of the 2010s) suggests the transition is real and disruptive.

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## **Part II — Digital Public Infrastructure: The India Stack Model**

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### **UPI and the Payments Revolution**

The Unified Payments Interface (UPI) — launched in 2016 by the National Payments Corporation of India (NPCI) — has become the most successful national digital payments system in history by transaction volume. In FY2025, UPI processed 131 billion transactions with a cumulative value of approximately \$2.3 trillion — more than India's annual GDP per capita multiplied by every household in the country. The system is real-time, interoperable across all banks, and costs merchants zero to accept.

The scale of UPI's penetration has effectively eliminated cash as the dominant payment mechanism for urban India and is rapidly doing so for rural India — the Jan Dhan Yojana programme has opened 520 million zero-balance bank accounts, extending UPI access to previously unbanked populations. The geopolitical dimension: India is actively exporting the UPI model to Southeast Asia (Singapore, UAE, France, Nepal, Sri Lanka have all activated UPI corridors), positioning Indian digital infrastructure standards as an alternative to Western payment rails (Visa/Mastercard) and Chinese systems (Alipay/WeChat Pay).

### **Aadhaar and the Identity Layer**

Aadhaar — India's biometric national ID system — has enrolled 1.38 billion individuals, making it the largest biometric database in human history. The system assigns a unique 12-digit number linked to fingerprints and iris scans, enabling paperless identity verification for government services, financial inclusion, telecom SIM provisioning, and increasingly, private sector KYC. Aadhaar's API-based architecture (eKYC, DigiLocker, eSign) forms the foundation of India Stack — the open API ecosystem that allows private companies to build digital services on top of government-provided identity, payments, and data infrastructure.

The India Stack model — Aadhaar + UPI + DigiLocker + ONDC (Open Network for Digital Commerce) + Account Aggregator — is the world's most ambitious attempt to build an open, interoperable digital economy infrastructure at national scale. At the G20 India Presidency summit (2023), Prime Minister Modi promoted India Stack as a global public good, proposing the framework be adopted by developing economies as an alternative to Big Tech-dominated digital ecosystems.

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## Part III — Semiconductors: The Upstream Ambition

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### India Semiconductor Mission

The Indian Semiconductor Mission (ISM) — launched in December 2021 with an incentive commitment of Rs 76,000 crore (approximately \$9.1 billion at then-rates, subsequently expanded to ~\$15 billion equivalent) — represents India's first serious attempt to develop a domestic semiconductor manufacturing capability. The mission offers approved fab projects up to 50% of project cost as capital subsidy from the central government, plus additional state government support, creating an effective 60-70% subsidy on greenfield fab construction.

Three major projects have received ISM approval:

Micron Technology (Sanand, Gujarat): A \$2.75 billion semiconductor assembly, testing, marking and packaging (ATMP) facility for DRAM and NAND Flash. First building commissioned in 2024; first India-made chips shipped in early 2025. Micron's India ATMP is a packaging operation, not a wafer fab — it does not produce chips from silicon but assembles and tests dies manufactured in TSMC/Samsung fabs. Nevertheless, it establishes India's first modern semiconductor manufacturing footprint and creates the supply chain ecosystem (chemicals, gases, equipment service) that a future fab would require.

Tata Electronics-Powerchip Semiconductor Manufacturing Corporation (PSMC) JV (Dholera, Gujarat): A \$11 billion wafer fabrication facility targeting 28nm-90nm process nodes. Powerchip (Taiwan) provides the technology transfer; Tata brings the capital and operational management. The fab will produce mature-node chips (display drivers, power management ICs, automotive chips) rather than advanced logic. Construction began 2024; production target 2026-2027. This is India's first true wafer fabrication facility and represents a historic milestone in India's semiconductor journey.

HCL-Foxconn JV (Jewar, Uttar Pradesh): An OSAT (outsourced semiconductor assembly and test) facility with a broader portfolio than Micron's single-company operation. Smaller scale but adds geographic diversification.

### The Gap Between Ambition and Capability

India's semiconductor programme faces structural constraints that cannot be resolved by capital alone. India has no domestic EDA (electronic design automation) software capability to compete with Synopsys and Cadence. India has no domestic semiconductor equipment industry; all lithography, etch, and deposition tools must be imported from ASML, Applied Materials, Lam Research, and KLA. India's chemical and materials supply chain for semiconductor-grade chemicals is nascent. The talent pipeline — while large in software engineering — has limited depth in process engineering, fab operations, and materials science at the level required for sub-5nm process development.

India's semiconductor ambition is thus realistic at mature nodes (28nm-90nm, producing chips for automotive, IoT, consumer electronics, and power management) but remains aspirational at advanced logic nodes (below 7nm) where TSMC, Samsung, and Intel Foundry are the only realistic near-term options globally.

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## Part IV — Space Technology and ISRO: The New Frontier

### Chandrayaan-3 and the South Pole Triumph

On August 23, 2023, India's Chandrayaan-3 lander Vikram touched down near the Moon's south pole — a region of intense scientific and strategic interest because of confirmed water ice deposits in permanently shadowed craters. India became the fourth nation (after the Soviet Union, United States, and China) to achieve a soft landing on the Moon, and the first to do so near the south pole. Russia's

Luna-25 mission, which crashed four days earlier attempting the same landing zone, underscored the difficulty of the feat.

The south pole achievement transformed India's international standing in the space domain. Within weeks, the US, Russia, China, and ESA had all reached out to India for lunar cooperation discussions. India's pragmatic positioning — a space power that is non-aligned, technologically capable, and willing to partner with any major space agency — makes ISRO a geopolitically flexible instrument for building science and commercial partnerships across the US-China divide.

## **IN-SPACE and the Commercial Space Era**

India's IN-SPACE (Indian National Space Promotion and Authorisation Centre) — established in 2020 — opened India's space sector to private companies for the first time in the programme's history. By 2026, over 250 Indian space startups have been incorporated; Skyroot Aerospace (Vikram-S rocket, India's first private rocket launch 2022), Agnikul Cosmos (Agnibaan small satellite launcher), and Bellatrix Aerospace (electric propulsion) represent a nascent but genuine private space industrial ecosystem.

India's target: achieve 9% of global space market share by 2030 (up from approximately 2-3% in 2025), with particular focus on small satellite launch services, remote sensing, and space-based communications for India's own growing satellite broadband ambitions (ISRO's GSAT Next-Generation programme and competition with Starlink for India's internet infrastructure).

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## **Conclusion: The Technology Superpower in Formation**

India's technology journey from software back-office to semiconductor ambition to digital public infrastructure architect spans three distinct developmental phases, all running simultaneously in 2026. The IT services industry — \$282 billion and structurally challenged by AI — must reinvent its value proposition or face a decade of margin compression. The semiconductor mission — \$15 billion committed, first fabs commissioning — is a 15-20 year project to build a real chip manufacturing ecosystem from near-zero, with realistic near-term ambitions (mature nodes) and aspirational long-term targets (advanced logic). The digital public infrastructure achievement — UPI, Aadhaar, India Stack — is already the world's most advanced public digital architecture and is being exported globally as India's most underrated soft-power asset.

The synthesis: India is building the technology base of a 21st-century great power — not through any single dramatic leap but through the simultaneous accumulation of capabilities across software, digital infrastructure, space, and now hardware manufacturing. The pace of accumulation is accelerating. The gap with China — which leads India by a decade or more in semiconductor sophistication — is narrowing in some dimensions (AI research, digital payments, space) while remaining wide in others (wafer fab, advanced manufacturing equipment). India's technology empire is real, growing, and strategically consequential.

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All data from NASSCOM, MEITY, NPCI, ISRO, UIDAI, and cited sources.